ML-Based Condition Monitoring and Remaining Life Prediction of Turbofan Engines

Names of all batchmates:

Madaka Shreyaj Naidu – 240962594

Advait Balachandar – 240962600

Dipen Batharaju – 240962574

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B.Tech. in CSE (AI & ML)

School of Computer Engineering

Manipal Institute of Technology

Manipal Academy of Higher Education

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Title of the Problem

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Abstract

Commercial aviation and industrial sectors heavily rely on the continuous and safe operation of turbofan engines. Unexpected engine failures not only incur high maintenance costs and operational downtime but also pose serious safety risks. Traditional maintenance strategies, such as run-to-failure or scheduled preventive maintenance, can lead to either unexpected breakdowns or premature replacement of healthy components. To address this real-world problem, this mini-project proposes a data-driven predictive maintenance framework for estimating the Remaining Useful Life (RUL) of turbofan engines. The NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset will be used to analyze multivariate time-series sensor data, including temperature, pressure, and vibration-related measurements recorded over multiple operational cycles. Machine learning regression models, including Random Forest, XGBoost, LightGBM, and Long Short-Term Memory (LSTM) networks, will be developed and compared for RUL prediction. The methodology includes data preprocessing, exploratory data analysis, feature engineering, RUL target construction, normalization, and sliding-window processing for temporal forecasting. Model performance will be evaluated primarily using RMSE and MAE. The project aims to support a transition from reactive maintenance to proactive maintenance by enabling earlier intervention, better component utilization, reduced maintenance wastage, and improved engine safety.

Keywords

Predictive Maintenance; Remaining Useful Life (RUL); Machine Learning; Time-Series Forecasting; NASA C-MAPSS; Turbofan Engine; Deep Learning

Introduction

Turbofan engines operate under demanding and changing conditions, making continuous condition monitoring important for safe and efficient operation. A useful maintenance system should not only identify degradation but also estimate how many operating cycles remain before failure. Remaining Useful Life prediction provides this capability by transforming historical sensor measurements into an estimate of future engine life. Machine learning is well suited to this task because engine degradation is represented by multivariate and time-dependent sensor patterns. In this project, the NASA C-MAPSS dataset is used to study run-to-failure engine behavior and build predictive models. The study focuses on preprocessing sensor data, identifying useful degradation-related features, constructing RUL targets, and comparing ensemble learning methods with an LSTM model. The expected outcome is a comparative predictive-maintenance framework that can estimate RUL accurately and help maintenance decisions become more proactive.

Literature Survey

Existing research demonstrates the effectiveness of data-driven and deep-learning approaches for turbofan RUL estimation. Studies have explored semi-supervised deep architectures, gradient boosting with explainability, CNN-LSTM models, attention-based networks, machine-learning prognostics, hybrid deep-learning architectures, and degradation modeling. Recent work also compares time-series forecasting methods and investigates RUL prediction under different operating conditions. These studies motivate the comparison of Random Forest, XGBoost, LightGBM, and LSTM models in the proposed work. The literature also supports the use of sliding windows, feature engineering, scaling, and RMSE/MAE-based evaluation for degradation forecasting.

Objectives

1. To design and implement machine learning regression models that estimate the Remaining Useful Life (RUL) of turbofan engines using multivariate time-series sensor data.

2. To perform exploratory data analysis and feature engineering to identify important degradation-related sensors across operating conditions.

3. To compare ensemble learning algorithms such as Random Forest, XGBoost, and LightGBM with an LSTM deep-learning architecture for run-to-failure RUL prediction.

Data Source

Dataset: NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) Dataset.
Format: Tabular multivariate time-series data (.txt / CSV).
Description: The dataset contains simulated run-to-failure degradation data for a fleet of turbofan engines under varying operating conditions and fault modes, with operational settings and 21 sensor measurements recorded per cycle. The dataset is the primary reference data source for model development and evaluation.

Methodology

1. Data Ingestion and Exploration: Load the NASA C-MAPSS dataset and perform EDA to visualize sensor trends, identify constant sensors, and inspect correlations.

2. Data Preprocessing and Target Definition: Construct the RUL target using a piece-wise linear degradation model, clip early RUL values at a maximum threshold, and normalize sensor readings using Min-Max scaling.

3. Feature Engineering: Use sliding-window techniques to capture temporal dependencies and sequential degradation patterns in multivariate sensor data.

4. Model Development: Develop baseline regression models and ensemble methods including Random Forest, XGBoost, and LightGBM, followed by an LSTM model for sequential forecasting.

5. Evaluation: Evaluate models on unseen test data using RMSE and MAE and compare predicted RUL with ground-truth RUL values.

Implementation Details

The implementation will be carried out using Python-based machine-learning and data-analysis tools. The workflow will include dataset loading, data cleaning, exploratory visualization, sensor selection, RUL-label generation, Min-Max normalization, sliding-window sequence generation, model training, hyperparameter tuning, and performance evaluation. Ensemble models will provide non-sequential baselines and the LSTM model will process temporal sequences. Results will be recorded using RMSE and MAE along with comparative visualizations of predicted and actual RUL.

Results and Discussion

The final results will present the predictive performance of Random Forest, XGBoost, LightGBM, and LSTM models on the NASA C-MAPSS test data. RMSE and MAE will be used as the principal quantitative measures. Comparative plots and tables will be used to identify the model that provides the most accurate and consistent RUL estimates. Discussion will focus on the effect of temporal feature engineering, sliding-window size, model architecture, and operating-condition variability on prediction performance.

Conclusion and Future Scope

The proposed project develops a machine-learning-based predictive maintenance framework for turbofan engines, with emphasis on Remaining Useful Life estimation. By combining sensor-data preprocessing, degradation-oriented feature engineering, ensemble learning, and LSTM-based sequence modeling, the work aims to provide a systematic comparison of RUL prediction approaches. Future work can extend the framework to additional operating conditions and fault modes, incorporate explainable-AI techniques, investigate more advanced temporal architectures, and explore deployment for near-real-time engine health monitoring.

Plan of Execution

• Finalization of synopsis, literature review, and dataset acquisition.

• Data preprocessing, handling missing/constant values, and exploratory data analysis.

• Feature engineering, formulation of the piece-wise RUL target, and data scaling.

• Implementation and tuning of baseline and ensemble algorithms.

• Development and training of LSTM models using sliding-window time-series formatting.

• Evaluation using RMSE and MAE followed by comparative analysis.

• Final documentation, visualization, report writing, and presentation preparation.

References

[1] A. L. Ellefsen, E. Bjørlykhaug, V. Æsøy, S. Shirzadi, and A. Ushakov, “Remaining useful life predictions for turbofan engine degradation using semi-supervised deep architecture,” Reliability Engineering & System Safety, vol. 183, pp. 240–251, 2019.

[2] X. Zhao and M. Jia, “Predictive maintenance of aircraft engines using gradient boosting and SHAP explainability,” Reliability Engineering & System Safety, vol. 198, p. 106900, 2020.

[3] Y. Wu, M. Yuan, S. Dong, L. Lin, and Y. Liu, “Remaining useful life estimation of turbofan engines using CNN-LSTM architecture,” IEEE Access, vol. 9, pp. 31448–31460, 2021.

[4] H. Huang, L. Huang, Y. Wang, and Y. Li, “Remaining useful life prediction for turbofan engines using an attention-based deep neural network,” IEEE Transactions on Reliability, vol. 70, no. 4, pp. 1642–1652, 2021.

[5] Z. Chen, M. Wu, R. Zhao, F. Geng, and Z. Zheng, “Machine learning based prognostics for remaining useful life prediction of turbofan engines,” Sensors, vol. 20, no. 21, p. 6061, 2020.

[6] Y. Liu, G. Zhao, and X. Wang, “A novel predictive maintenance framework for turbofan engines based on XGBoost and LSTM,” Applied Soft Computing, vol. 114, p. 108078, 2022.

[7] B. Wang, Y. Lei, N. Li, and N. Li, “Data-driven remaining useful life prediction using multi-scale deep feature extraction,” Mechanical Systems and Signal Processing, vol. 185, p. 109786, 2023.

[8] X. Li, Q. Ding, and J. Q. Sun, “An integrated machine learning approach for RUL estimation under multiple operating conditions,” ISA Transactions, vol. 121, pp. 351–365, 2022.

[9] A. Gupta, A. Singh, and S. Sharma, “Turbofan engine degradation modeling and RUL prediction using Random Forest and LightGBM,” Journal of Manufacturing Systems, vol. 68, pp. 120–132, 2023.

[10] D. Zhou, Z. Li, and Y. Wang, “Remaining useful life prediction of NASA C-MAPSS dataset using deep learning,” in Proc. IEEE International Conference on Prognostics and Health Management (ICPHM), pp. 1–6, 2021.

[11] M. Kim, J. Lee, and S. Park, “Time-series forecasting for aircraft engine predictive maintenance: A comparative study,” Expert Systems with Applications, vol. 237, p. 121478, 2024.

[12] M. Rahman and A. Faisal, “Evaluating machine learning models for run-to-failure degradation in complex systems,” IEEE Transactions on Industrial Informatics, vol. 18, no. 5, pp. 3245–3255, 2022.

[13] A. Al-Dulaimi, S. Zhai, and X. Li, “A hybrid deep learning architecture for remaining useful life prediction of turbofan engines,” Computers in Industry, vol. 129, p. 103445, 2021.

[14] J. Shi, H. Liu, and Y. Zhang, “Piecewise linear degradation modeling and RUL prediction for predictive maintenance,” Measurement, vol. 208, p. 112463, 2023.

[15] C. Zhang, P. Lim, A. K. Qin, and K. C. Tan, “A novel deep learning framework for remaining useful life prediction of machinery,” Mechanical Systems and Signal Processing, vol. 139, p. 106622, 2020.

[16] J. Li, X. Li, and D. He, “Data-driven approach for predicting remaining useful life of aircraft engines,” IEEE Access, vol. 7, pp. 33502–33512, 2019.

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